

CSDS 302 Discrete Math | Fall 2026 | Homework 1

Instructions

Section 1.2: 3abc(4 points each part), 6, 7aef(4 points each part), 9cg(4 points each part), 10bd(4 points each part), 12ac(4 points each part)

Section 1.3: 2abcd(4 points each part), 8, 13, 14, 15de(4 points each part), 18, 20

Section 7.1: 2abcdef(2 points each part), 4b(4 points each part)

Section 7.2: 7b(4 points each part), 8ab(4 points each part)

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Feedback

Score: 79/100

Section 1.2

Q6: -4

Section 1.3

2d: -1, missing (2,2)

13a: -1, should be $\{-1, 0, 1\}$ (integer), can't be continuous set

15: -4

18: -4

20: -4

Section 7.3:

7b: -1, it's not onto

8: -2, doesn't answer if it's onto

Zelan Eroz Espanto (Zere3)

Homework 1

Exercise 1.2

1. a. $A = \{1, 2\}$
 b. How many elements are in the set $\{1, 2, 3, 5\}$?
 c. How many elements are in the set $\{1, 2, 3, 5\}$?

2. a. NO
 b. YES
 c. YES

3. a. YES
 b. YES

4. a. YES
 b. YES

5. a. YES
 b. YES

6. a. YES
 b. YES

7. a. YES
 b. YES

8. a. YES
 b. YES

9. a. YES
 b. YES

10. a. YES
 b. YES

11. a. YES
 b. YES

12. a. YES
 b. YES

13. a. YES
 b. YES

14. a. YES
 b. YES

15. a. YES
 b. YES

16. a. YES
 b. YES

17. a. YES
 b. YES

18. a. YES
 b. YES

19. a. YES
 b. YES

20. a. YES
 b. YES

21. a. YES
 b. YES

22. a. YES
 b. YES

23. a. YES
 b. YES

24. a. YES
 b. YES

25. a. YES
 b. YES

26. a. YES
 b. YES

27. a. YES
 b. YES

28. a. YES
 b. YES

29. a. YES
 b. YES

30. a. YES
 b. YES

31. a. YES
 b. YES

32. a. YES
 b. YES

33. a. YES
 b. YES

34. a. YES
 b. YES

35. a. YES
 b. YES

36. a. YES
 b. YES

37. a. YES
 b. YES

38. a. YES
 b. YES

39. a. YES
 b. YES

40. a. YES
 b. YES

13. a. Domain: $f: 1 \leq x \leq 13$ Co-domain: $f: t, u, v, w$
 b. $f(1) = u$; $f(13) = w$; $f(1) = u$
 14. a. Domain: $f: 1, 2, 3, 4$ Co-domain: $f: a, b, c, d$
 b. $G(1) = G(2) = G(3) = G(4) = c$
 15. d.
 e.

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Exercise 7.1

2. a. Domain: $x = 1, 2, 3, 4$ Co-domain: $y = 1, 2, 3, 4, 5$
 b. $g(1), g(2), g(3) = b$
 c. Range: $\{b\}$
 d. No. Yes.
 e. $1, 2, 3, 4, 5$
 3. a. Find all functions from $X = \{a, b, c\}$ to $Y = \{u, v\}$.

Exercise 7.2

7. b. G is not one-to-one because $G(a) = G(b) = f$.
 G is onto because for every element in the co-domain when $x = a, b, c$ there is a corresponding x in X .
 8. a. No because $H(b) = H(c) = f$. It's not onto either because e and g in Y don't correspond to any element in X .
 b. Yes

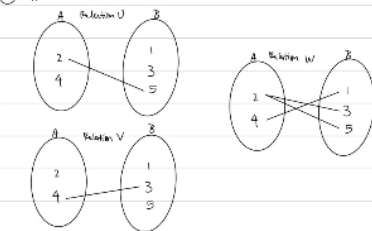
Exercise 1.3

2. a. $\frac{1}{2} - \frac{1}{3} = 0$; $2 \cdot 2 = 4$
 $\frac{1}{2} + \frac{1}{3} = 0$; $-1 \cdot 1 = -1$
 $\frac{1}{2} - \frac{1}{3} = 0$; $(3, 2) \in S$
 b. $S = \{(1, 3), (-2, 2), (-1, 1), (4, 1), (1, 1), (2, 2), (3, 3), (-2, 2)\}$

- c. $x \in C$ such that there exists at least one $y \in D$ making $(x, y) \in S$.
 Co-domain: All $y \in D$ such that there exists at least one $x \in C$ making $(x, y) \in S$.



8. $A = \{1, 2, 3, 4\}$ $B = \{1, 3, 5\}$



je t'aima co much zelan!!
 love your enemy too